

# 17.2 the STM32N6 Microcontroller: Enabling Intelligent Edge AI for IoT and Beyond

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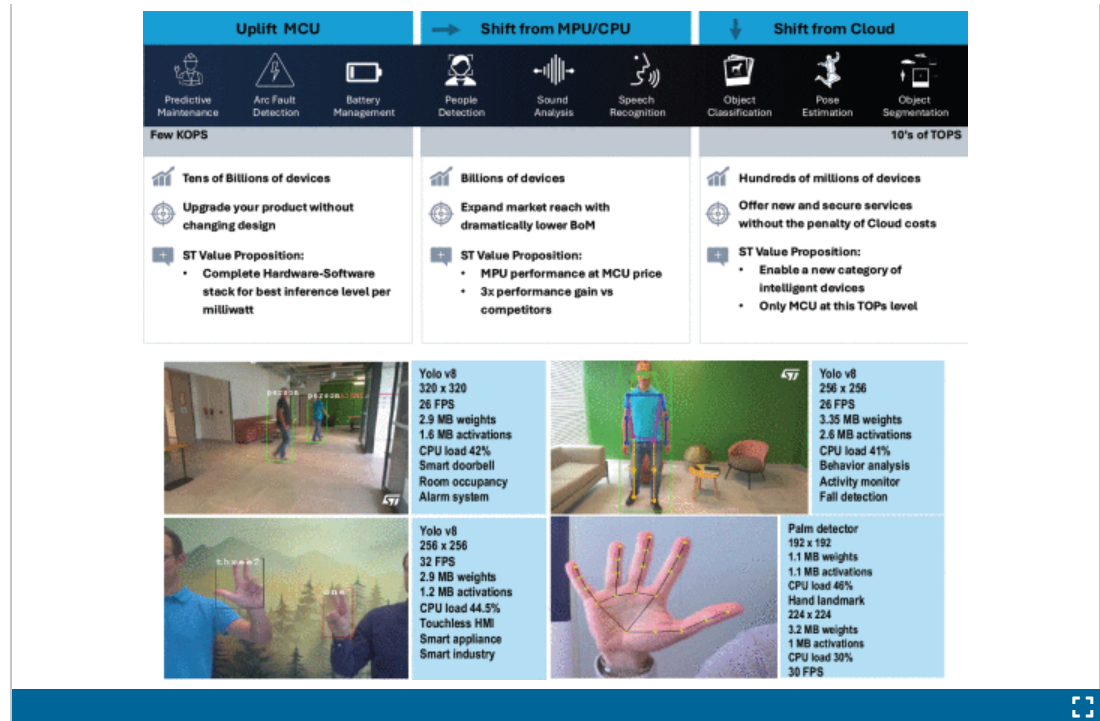
Contents

The STM32N6 microcontroller addresses the increasing demand for intelligent edge devices in IoT, wearable technology, industrial automation, and smart-home systems. It enables real-time, energy-efficient AI processing at the edge, reducing latency and enhancing data privacy by minimizing cloud dependency. At its core is the Neural-ART neural processing unit (NPU), a dedicated accelerator that supports complex AI inference with minimal power consumption. This paper explores the chip design, implementation in 16 nm FinFET, and performance, highlighting its ability to balance computational power and energy efficiency.

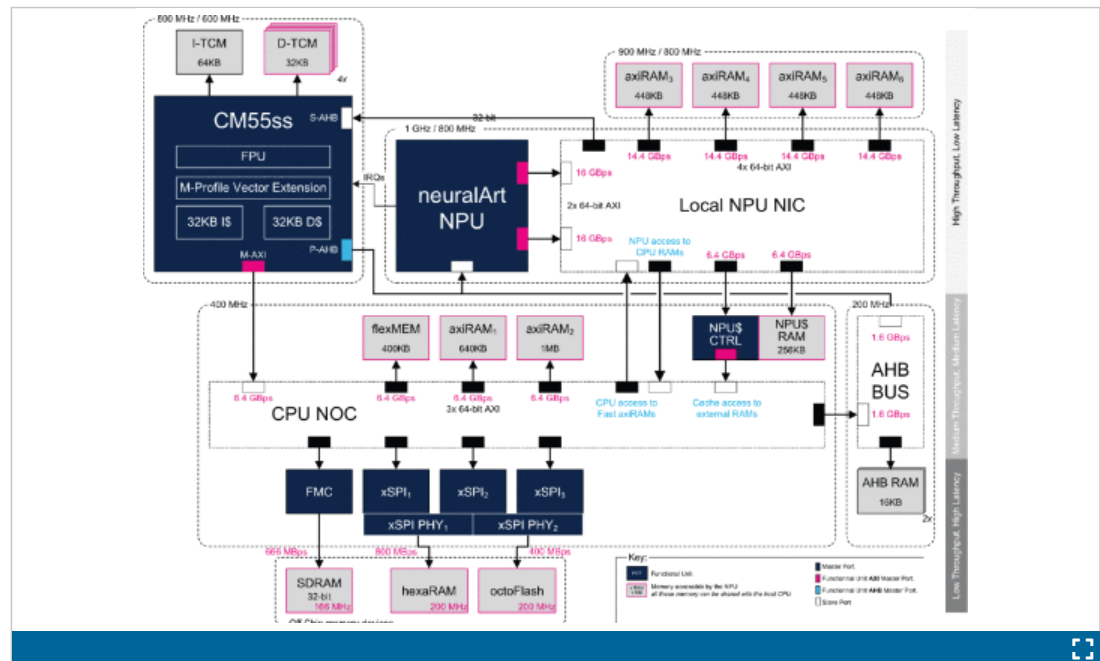
The rapid growth of embedded AI applications in IoT, wearables, and smart systems has increased the demand for energy-efficient, low-latency neural network processors. Traditional microcontrollers often fail to meet the processing requirements of modern algorithms. Consequently, dedicated hardware accelerators have emerged as a viable alternative [1]. Current methods often compromise adaptability for performance by relying on rigid-function components or reducing processing speed to accommodate varied kernel dimensions and layer arrangements.

The STM32N6 chips are based on an Arm® Cortex®-M55 core running at up to 800 MHz, while the NPU runs at up to 1 GHz, delivering a peak of 600 giga-operations per second (GOPS) and up to three tera-operations per second per watt (TOPS/W). Other coprocessors include a Chrom-ART accelerator for 2D graphics, a Chrom-GRC graphics resource cutter for round and other non-square displays, a 2.5D NeoChrom graphics accelerator, an H. 264 video encoder capable of 1080p15 or 720p30, and an image signal processor (ISP) targeting a 5 Mpixel camera at 30 frames per second (Fig. 17.2.2).





**Figure 17.2.1:**  
Edge AI macro-trends and STM32N6 application examples.



**Figure 17.2.2:**  
STM32N6 core architecture diagram, the soc architecture is built around a noc backbone connecting cpu, memories and peripherals and a fully connected nic supporting the npu and four memories plus the npu cache, the npu and the bus are connected via asynchronous links.

The chip offers a memory configuration with 4.2 MB of contiguous embedded RAM and 64+128 KB tightly coupled memories (TCMs), plus external interfaces for multiple memory types, including pseudo-static RAM (PSRAM), synchronous dynamic RAM (SDRAM), and both NOR and NAND flash. It also includes a wide range of external interfaces, such as Gigabit Ethernet and CSI2.0. An Arm® TrustZone® security subsystem is complemented by ST's proprietary SoC security architecture, side-channel-attack-resistant AES acceleration, tenant-aware firewalling, and the goal of achieving SESIP Level 3 and PSA Level 3 security certifications. Designed for edge AI applications, the STM32N6 combines low power consumption, advanced processing capacity, and a feature set tailored to industrial, consumer, and some automotive uses.

The STM32N6 is powered by the Neural-ART NPU, a reconfigurable NN processor designed to address these constraints by way of a streaming data-flow-based NPU that features dynamic kernel configuration, compression, and batch processing, enabling adaptive resource allocation for diverse workloads. A reconfigurable data transfer fabric integrated into the NPU and linked to the on-chip interconnect enhances data reuse and minimizes access to on/off-chip memory, a significant constraint in edge devices [3]. A low-power Arm® Cortex®-M55 microprocessor core with a suite of peripherals and interfaces supports simultaneous processing of data acquisition and transfer, pre- and postprocessing, as well as non-hardware-accelerated tasks. The Arm's Helium vector extensions boost machine learning workloads while running on the microcontroller core, reducing idle periods and improving overall system performance.

Recent advances in hardware acceleration for neural networks (NN) focus on optimizing energy efficiency and computational throughput for edge devices. Both research and industrial applications demonstrate that NNs are effective for many sophisticated functionalities; however, their high computational requirements necessitate dedicated hardware. Later designs [1] show that NNs can achieve high accuracy, but their substantial energy demands can make them impractical for embedded applications. In response, several energy-saving accelerators have been proposed. However, these designs often lack flexibility in handling varying kernel sizes and layer configurations. Recent architectures enhance memory efficiency by employing on-chip storage, yet many remain unsuitable for edge devices with limited resources. Conversely, microcontroller-based approaches prioritize energy efficiency but frequently compromise computational capability.

The NPU differentiates itself from previous designs by integrating dynamically configurable acceleration with autonomous layer processing. Unlike rigid ASIC-based architectures [3], this approach dynamically adjusts to the demands of individual layers without increasing power consumption. This adaptability is particularly beneficial for embedded applications, where minimizing latency and maximizing energy efficiency are critical [2]. Consumer and industrial AI algorithms require deeper topologies with many layers, millions of parameters, and varying kernel sizes, resulting in escalating bandwidth, power, and area costs. Meeting these constraints for embedded devices and applications requires efficiency through a hierarchical memory system and effective reuse of local data. In the STM32N6, the NPU relies on unidirectional links transporting data streams via a configurable fully connected switch, including DMAs and accelerators supporting convolutions, pooling, activation, scalar, and other processing kernels. This fabric allows the definition of an arbitrary number of concurrent, virtual processing chains at runtime. A full-featured backpressure mechanism handles data flow control and stream multicasting, enabling the reuse of a data stream at multiple endpoints. Linked lists control the fully autonomous processing of an entire epoch, which can be a fused combination of multiple layers. Multiple accelerators grouped or chained together handle varying feature map sizes and multiple kernels in parallel. The switch topology is defined at compile time and dynamically configured at runtime, with each node in the fabric graph representing a processing element or memory streaming channel.

As illustrated in Fig. 17.2.3 and Fig. 17.2.4, the design supports pipelined operation across multiple tiers while preserving worst-case latency constraints. The configurable accelerator framework (CAF) manages data movement and computation scheduling while automatically adjusting to diverse network layer topologies.



**Figure 17.2.3:**

The Neural-ART streaming dataflow architecture is a design time parametric NPU built around a streaming switch that can be defined with a mix of different accelerators, features and cardinalities to meet specific products requirements.



**Figure 17.2.4:**

The neural-art maximizes local data reuse matching bandwidth, buffer space, mac units, to accelerators that can be rearranged to balance bw and computing for different layer topologies. iterations on input/output channels can handle arbitrary tensor and kernel dimensions.

The NPU supports both INT8 and INT16 data formats and dynamically adjusts to variable workloads, reduces memory operations, and minimizes delays, making it particularly suitable for embedded AI implementations.

This design-time parametric architecture represents a step toward a paradigm shift from the Von Neumann architecture to a flexible, dedicated dataflow stream processing engine. The instance integrated into the STM32N6 seamlessly connects to the MCU (Microcontroller Unit) backbone via two 64b AXI interfaces and associated DMA streaming channels, while also supporting direct streaming from the imaging sensor pipeline and a NPU data cache for both weights and activations.

A notable feature is the advanced stream ciphering/deciphering capability, which protects the intellectual property of application deployment for both kernel weights and activations without noticeable performance degradation.

The chip is designed in TSMC 16FF (16nm) with a die size of 17.5 mm<sup>2</sup> (Fig. 17.2.7). Fine and coarse clock gating is applied at the logic and processing element level, reducing dynamic inference power per layer and during idle periods. Architectural innovations, such as asynchronous bus nodes, high-end memories, and a network-on-chip backbone, combined with circuit-level optimizations, result in a balanced design suitable for energyconstrained edge deployments.

Figure 17.2.5 illustrates the execution of a YOLOv5 algorithm commonly used in the industry for image object detection and classification. It details performance and power consumption metrics when all layers are mapped onto the NPU. The top charts display the average power consumption of the AI subsystem (NPU and memories) and the execution time for hardware epochs generated by the mapper tool to complete an inference run. The ratio of epoch execution time to power depends on the mapping efficiency of each layer, considering its parameters, hardware capabilities, and memory bandwidth requirements. The network achieves over 100 frames per second at maximum chip performance, enabling reduced frequency and supply voltage in the field to meet a 30 frames per second real-time target. The bottom three charts in Fig. 17.2.5 show the relative power distribution across the AI subsystem for the NPU, bus, and memories, the NPU's internal power breakdown, including weights and activation buffers, and its subunits.



**Figure 17.2.5:**

PicoYolo v5 INT8 inf time 10.59 ms, average AI subsystem power (NPU + memories) 94mW (1.51 TOPS/W), top/middle: average power and compute time for each epoch/ fused layers, bottom left/right, power break down for the whole AI subsystem (NPU, Bus, Mems), NPU+buffers, NPU logic.

Edge AI with integrated NPUs delivers groundbreaking capabilities to IoT, wearables, healthcare, industrial automation, and smart home systems by providing powerful, lowlatency intelligence directly on the device. For the first time, tiny microcontrollers can perform complex computer vision tasks that were once limited to large, costly, and powerhungry microprocessors (MPUs). This breakthrough not only enhances the user's experience of existing products but also enables the development of entirely new solutions that were previously beyond technical and commercial reach. With the help of integrated efficient NPUs, MCUs can now handle diverse vision tasks such as object detection, gesture recognition, and quality inspection, enabling innovative and cost-effective solutions across numerous industries and use cases. In the wearables domain, where a small footprint and low power consumption are crucial, smart glasses can perform realtime analytics on camera images and deliver personalized experiences to users. Smart access control systems, which rely on reliable face recognition algorithms, require high processing power to ensure smooth user experience. Since these systems need to be cost-effective for widespread adoption, MCU-based solutions are preferred. The same applies to automatic license plate recognition systems, which have traditionally been offered only by expensive solutions. STM32 MCUs have been widely adopted by drone manufacturers for flight controllers. With the STM32N6, flight controllers will be equipped with computer vision capabilities to enable features such as follow-me and obstacle avoidance. Figure 17.2.1 shows snapshots of inference running on the device for four different applications, including people detection and pose estimation, hand gesture and pose recognition. Applications like these were previously possible only with much more powerful and costly devices.

In-memory computing (IMC) [6] significantly reduces data transfer with memory, hence power consumption. We are in a mature stage of prototyping this technology for both digital and analog versions, recording up to 8.3× and 16.7× TOPS/W improvements for digital and analog IMC, respectively ( Fig.

17.2.6). These technologies support advanced quantization at 1,2,4, and 8 bits for further performance improvements and model size reduction while ensuring seamless workflow integration in the continuity of the current NPU.



**Figure 17.2.6:** STMicroelectronics roadmap for non-von neumann NPUs relies on SRAM-based 1-2-4-8b in-memory computing multi-tiled NN accelerators [2], up to 8.3× and 16.7× for digital and analog (approximate) performance increase over STM32N6 peak efficiency measured of 3 TOPS/W in INT8 precision.



**Figure 17.2.7:** STM32N6 die photo.

The STM32N6’s adaptability to multiple use cases, varying AI algorithm functionality, and NPU autonomous layer processing capabilities make it particularly suitable for real-time applications in industrial IoT, wearables, and smart agriculture. Its alignment with widely adopted ML frameworks ensures real-world applicability, effectively bridging high-speed acceleration with the limitations of edge devices.

**ACKNOWLEDGMENT**

The STM32N6 family is the result of the coordinated effort of many STMicroelectronics engineers and experts, that couldn't be explicitly mentioned but are all deserving of recognition.